

Understanding Inequalities Worksheet

Fundamentals:

1 Circle the **true** statements:

$5 < 7$

$11 \leq 5$

$12 \neq 4$

$7 \geq 4$

$6 = 2$

$17 \geq 7$

$3 > 3$

$15 < 5$

$6 \geq 6$

$7 = 7$

$1 \leq 1$

$5 \neq 5$

2 Write down the meaning of the following statements in words:

$6 < 9$

Answer _____

$7 \geq 4$

Answer _____

$3 \neq 8$

Answer _____

$1 < 4 < 5$

Answer _____

$2 \leq 8 \leq 11$

Answer _____

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Intermediate:

1 Write down the integer (whole number) values of x in each case:

$$0 < x < 4$$

Answer _____

$$1 \leq x \leq 6$$

Answer _____

$$3 \leq x < 8$$

Answer _____

$$1 < x < 5 \quad \text{and} \quad x \neq 3$$

Answer _____

2 Write down an inequality that matches the following written statements:

- Seven is greater than three.

Answer _____

- Eight is greater than five and less than eleven.

Answer _____

- Sixteen is greater than, or equal to x .

Answer _____

- Seventeen is not equal to y .

Answer _____

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Challenge:

1 Given the following:

$$x < 2 \quad \text{and} \quad y > 4$$

Circle the statement(s) which are **always true**:

$y < x$

$y > x$

$y \geq x$

$y \leq x$

2 Given the following:

$$x < 7 \quad \text{and} \quad y > 5$$

Circle the statement(s) which are **always true**:

$y < x$

$y > x$

$y \geq x$

$y \leq x$